

COURSE-END PROJECT 1

Implementing SSH Brute-Force Simulation, Log Analysis, and Defensive Hardening on Windows

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Isolated VMware Security Lab

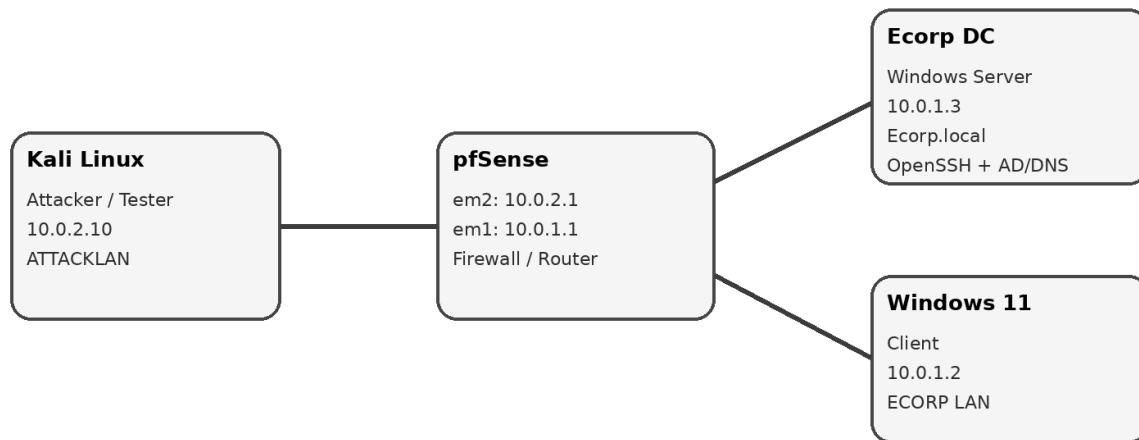


Figure 1: Isolated VMware lab topology and addressing.

1. Executive Summary

This report documents the supplied Course-End Project 2 scenario: a controlled SSH brute-force simulation against Windows OpenSSH, followed by log analysis, successful-authentication validation, SSH hardening, and public-key authentication.

- Hydra SSH password-guessing simulation against 10.0.1.3.
- OpenSSH/Operational failure analysis from source 10.0.2.10.
- Successful sstest authentication and Windows Security Event ID 4624.
- AllowGroups and Windows Defender Firewall access restrictions.
- Ed25519 public-key authentication through authorized_keys.

2. Project Objective and Required Tasks

The supplied project brief requires the following eight tasks:

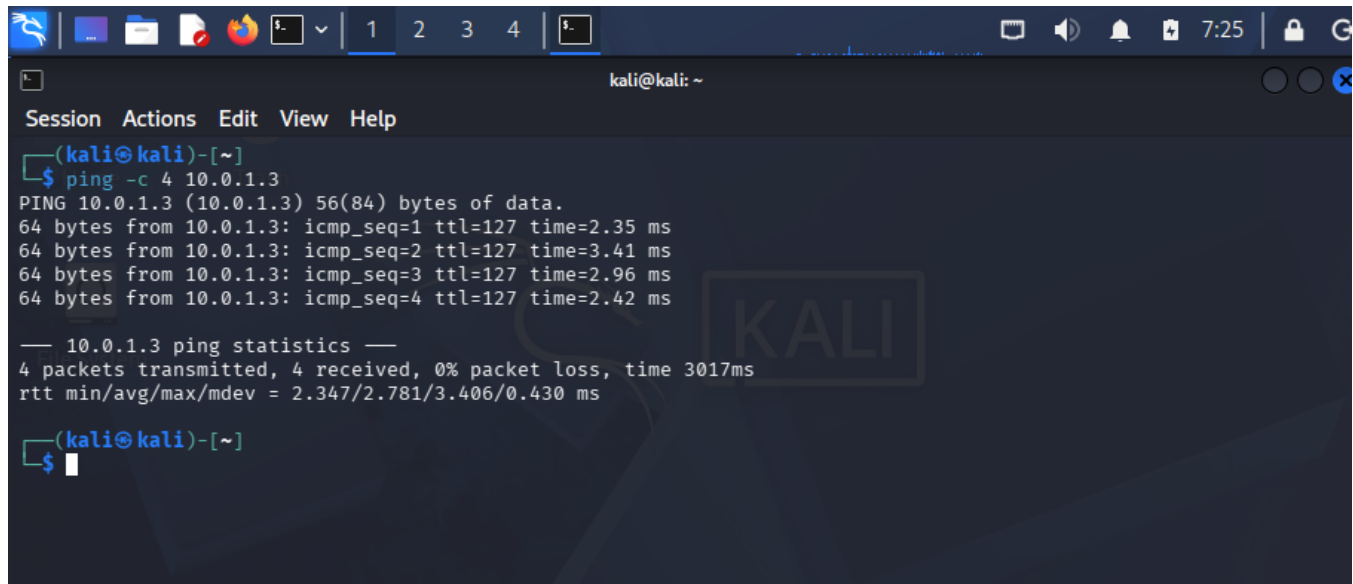
- Task 1: Verify network connectivity between attacker and victim systems
- Task 2: Install and configure the OpenSSH server on the Windows target
- Task 3: Create test user accounts and verify baseline SSH authentication
- Task 4: Simulate an SSH brute-force attack from Kali Linux
- Task 5: Investigate SSH authentication failures using Windows Security logs
- Task 6: Create and validate successful authentication events for compromise assessment
- Task 7: Implement SSH hardening and access control restrictions
- Task 8: Implement and validate certificate-based SSH authentication

3. Lab Environment

- Kali Linux — Attacker / Tester — 10.0.2.10 — ATTACKLAN
- pfSense — Firewall / Router
- Ecorp DC — Windows Server, AD DS/DNS, OpenSSH — 10.0.1.3 — Ecorp.local
- Windows 11 — Domain client — 10.0.1.2 — ECORP LAN

4. Task 1 — Network Connectivity

Kali 10.0.2.10 and the Windows Ecorp DC 10.0.1.3 were connected through pfSense. Connectivity was verified before SSH testing.

A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~'. The terminal shows a command prompt where the user has entered 'ping -c 4 10.0.1.3'. The output shows four successful ping responses from 10.0.1.3 with varying times (2.35 ms, 3.41 ms, 2.96 ms, 2.42 ms). Below the ping results, it shows '10.0.1.3 ping statistics' with '4 packets transmitted, 4 received, 0% packet loss, time 3017ms' and 'rtt min/avg/max/mdev = 2.347/2.781/3.406/0.430 ms'. The terminal prompt is now ready for the next command.

```
kali@kali: ~  
Session Actions Edit View Help  
(kali@kali)-[~]  
$ ping -c 4 10.0.1.3  
PING 10.0.1.3 (10.0.1.3) 56(84) bytes of data:  
64 bytes from 10.0.1.3: icmp_seq=1 ttl=127 time=2.35 ms  
64 bytes from 10.0.1.3: icmp_seq=2 ttl=127 time=3.41 ms  
64 bytes from 10.0.1.3: icmp_seq=3 ttl=127 time=2.96 ms  
64 bytes from 10.0.1.3: icmp_seq=4 ttl=127 time=2.42 ms  
  
— 10.0.1.3 ping statistics —  
4 packets transmitted, 4 received, 0% packet loss, time 3017ms  
rtt min/avg/max/mdev = 2.347/2.781/3.406/0.430 ms  
  
(kali@kali)-[~]  
$
```

Figure 1: Connection Check.

5. Task 2 — OpenSSH Installation and Configuration

Objective

The objective of this task was to install and configure the OpenSSH Server on the Windows Server and verify that the SSH service was running correctly.

Implementation

OpenSSH Server was configured on the Windows Server (Ecorp DC). The SSH service (sshd) was started using PowerShell and configured to accept SSH connections.

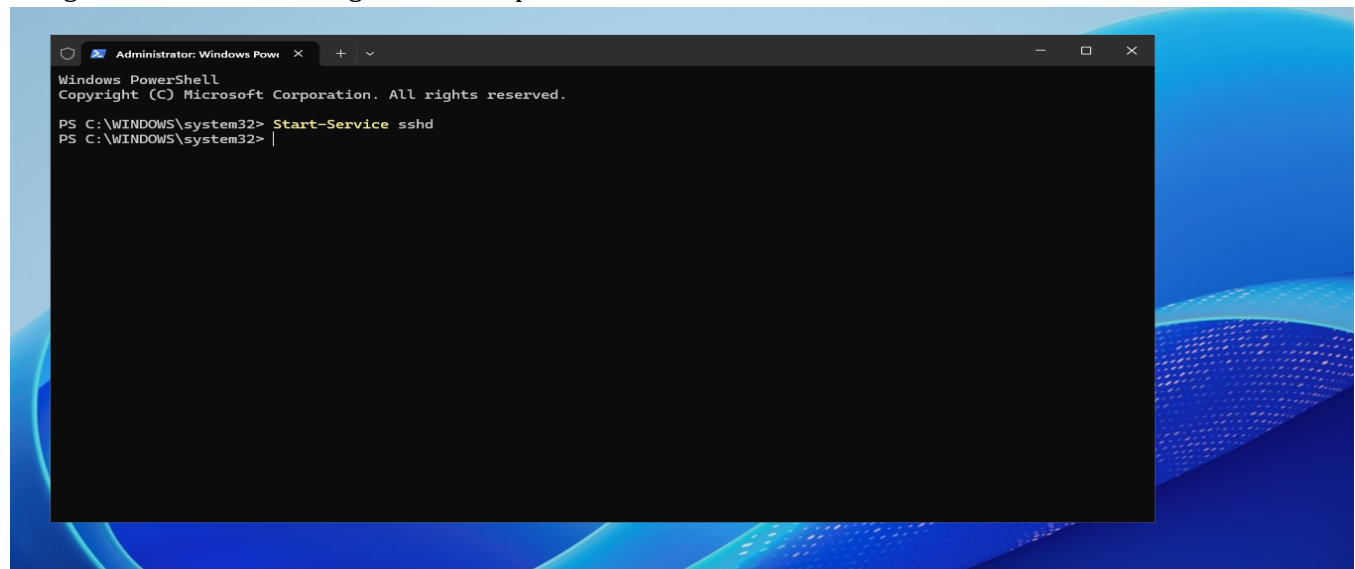
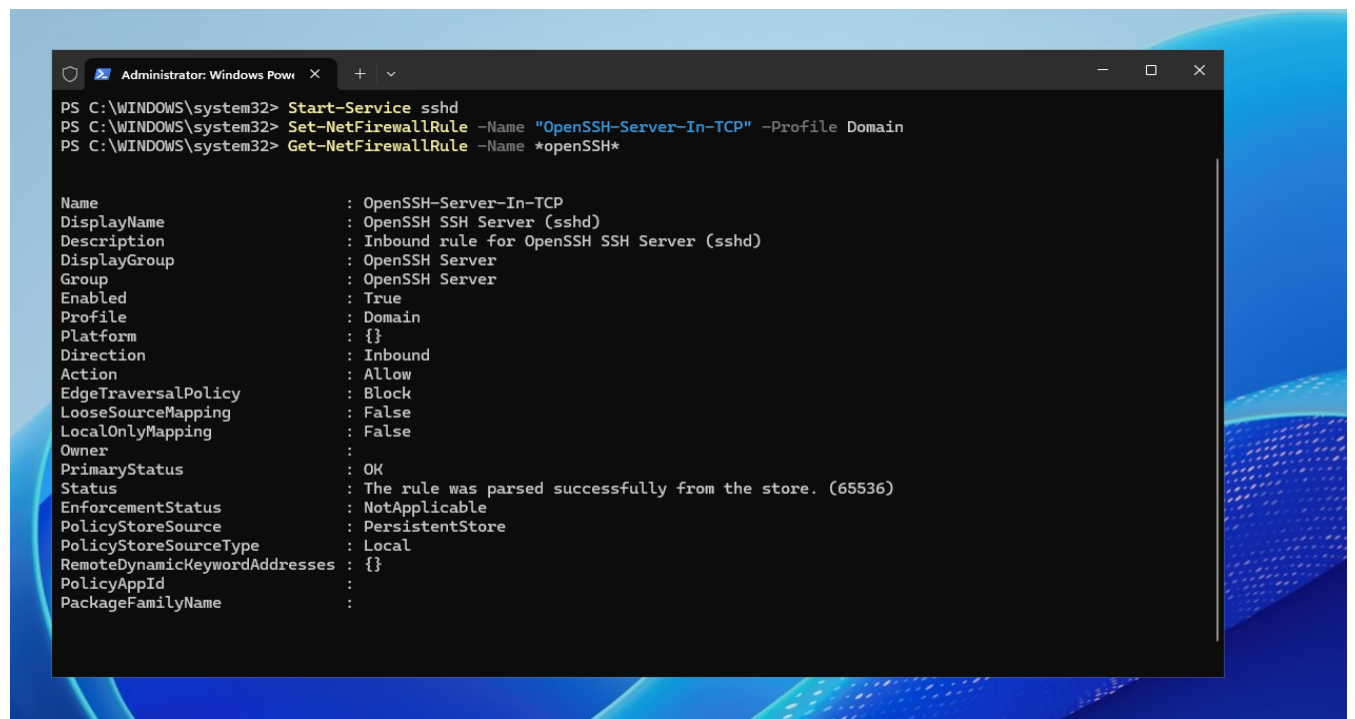


Figure 2: Start OpenSSH.



6. Task 3 — Test User and Baseline SSH Authentication

```
ssh sstest@10.0.1.3  
whoami
```

The authorized test account sstest successfully authenticated and was validated as ecorp\sstest.

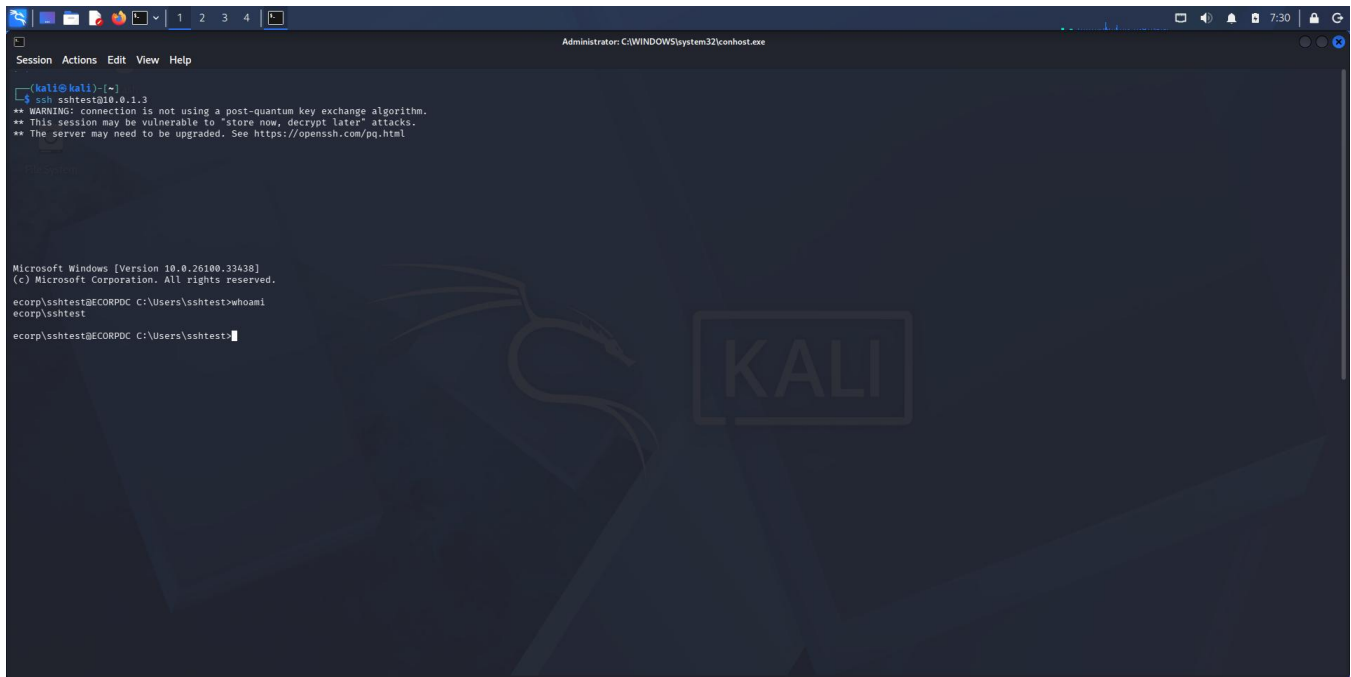
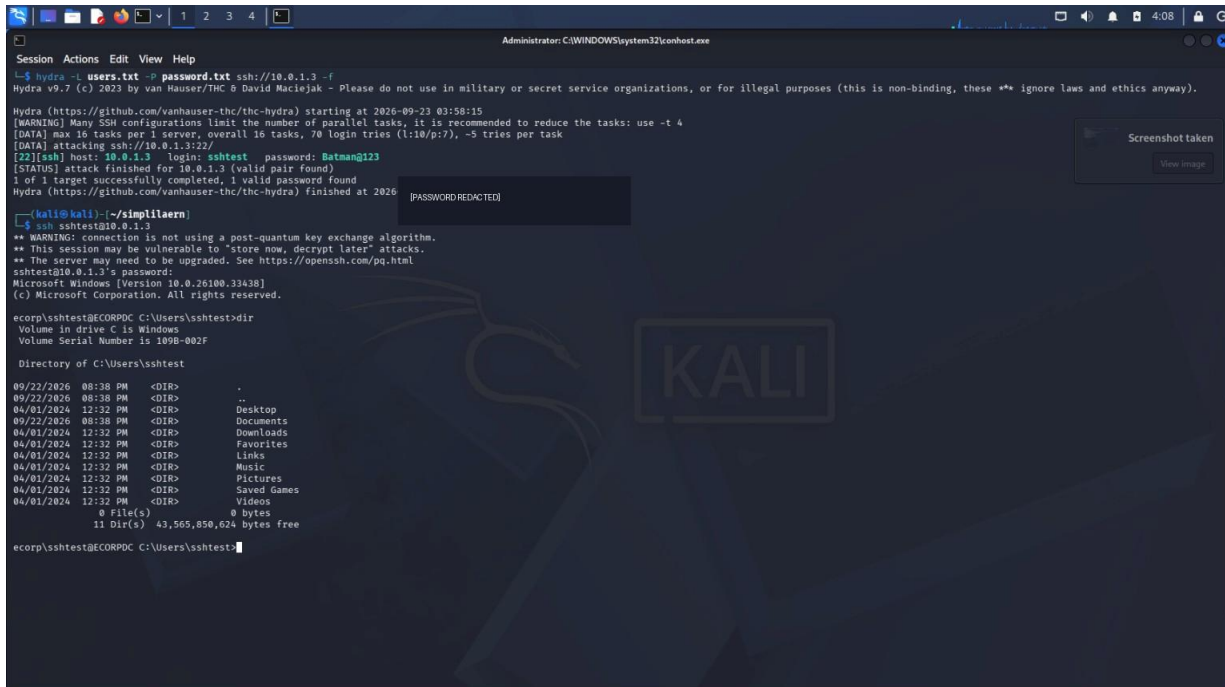


Figure 3: Baseline SSH session and whoami validation.

7. Task 4 — Controlled SSH Brute-Force Simulation

```
hydra -L users.txt -P password.txt ssh://10.0.1.3 -f
```

Hydra was used only inside the isolated lab. The credential shown in the original screenshot has been redacted.



```
Administrator: C:\WINDOWS\system32\cmd.exe
Session Actions Edit View Help
$ hydra -L users.txt -P password.txt ssh://10.0.1.3 -f
Hydra v9.7 (c) 2021 by van Hauser/THC & David Maciejak - Please do not use in military or secret service organizations, or for illegal purposes (this is non-binding, these ** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-09-23 03:58:15
[WARNING] Many SSH configurations limit the number of parallel tasks, it is recommended to reduce the tasks: use -t 4
[DATA] max 16 tasks per 1 server, overall 16 tasks, 70 login tries (1:10/p:7), ~5 tries per task
[DATA] attacking ssh://10.0.1.3:22/
[22][ssh] host: 10.0.1.3 login: sshtest password: Batman@123
[STATUS] attack finished for 10.0.1.3 (valid pair found)
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-09-23 03:58:15

kali@kali:~/simplilaern$ ssh sshtest@10.0.1.3
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
sshtest@10.0.1.3's password:
Microsoft Windows [Version 10.0.26100.33438]
(c) Microsoft Corporation. All rights reserved.

ecorp\sshtest@ECORPDC C:\Users\sshtest>dir
Volume in drive C is Windows
Volume Serial Number is 109B-002F

Directory of C:\Users\sshtest

09/22/2026  08:38 PM    <DIR>          .
09/22/2026  08:38 PM    <DIR>          ..
04/01/2024  12:32 PM    <DIR>          Desktop
09/22/2026  08:38 PM    <DIR>          Documents
04/01/2024  12:32 PM    <DIR>          Downloads
04/01/2024  12:32 PM    <DIR>          Favorites
04/01/2024  12:32 PM    <DIR>          Links
04/01/2024  12:32 PM    <DIR>          Music
04/01/2024  12:32 PM    <DIR>          Pictures
04/01/2024  12:32 PM    <DIR>          Saved Games
04/01/2024  12:32 PM    <DIR>          Videos
             0 File(s)              0 bytes
             11 Dir(s)  43,565,850,624 bytes free

ecorp\sshtest@ECORPDC C:\Users\sshtest>
```

Figure 4: Hydra brute-force simulation result (credential redacted).

8. Task 5 — SSH Authentication Failure Log Analysis

```
Get-WinEvent -LogName "OpenSSH/Operational" -MaxEvents 200 |  
Where-Object {$_.Message -match "failed|failure|invalid|password|preauth"} |  
Select-Object TimeCreated, Id, Message | Format-List
```

The logs recorded failed passwords, invalid usernames, pre-authentication disconnects, and source address 10.0.2.10.

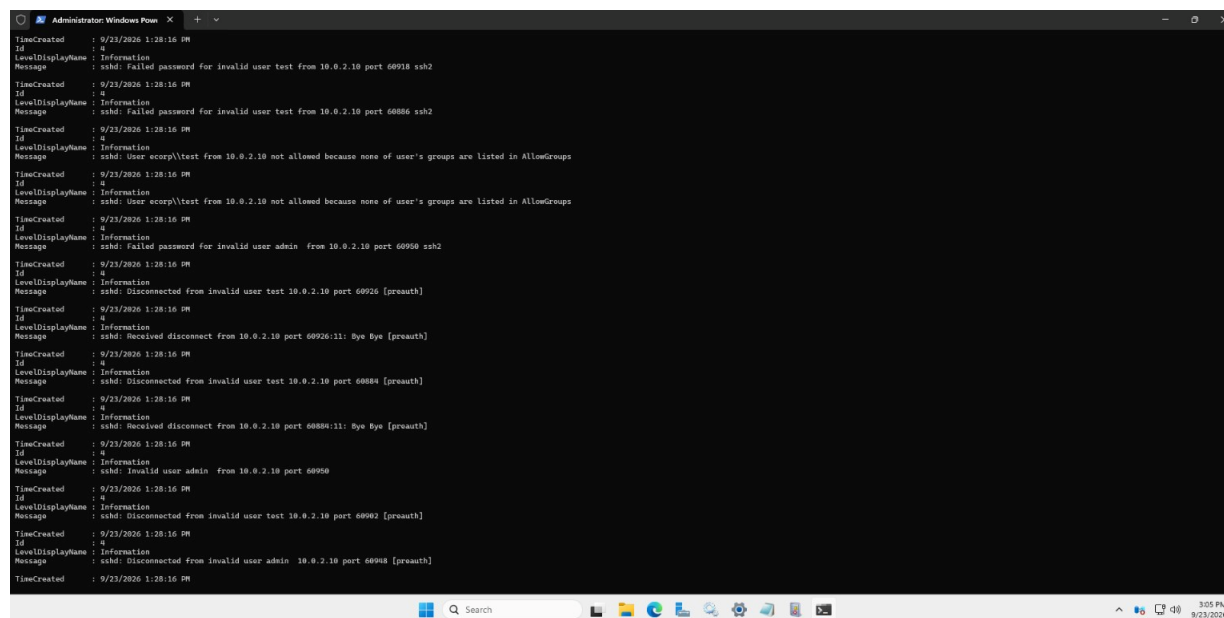


Figure 5: OpenSSH failure log analysis.

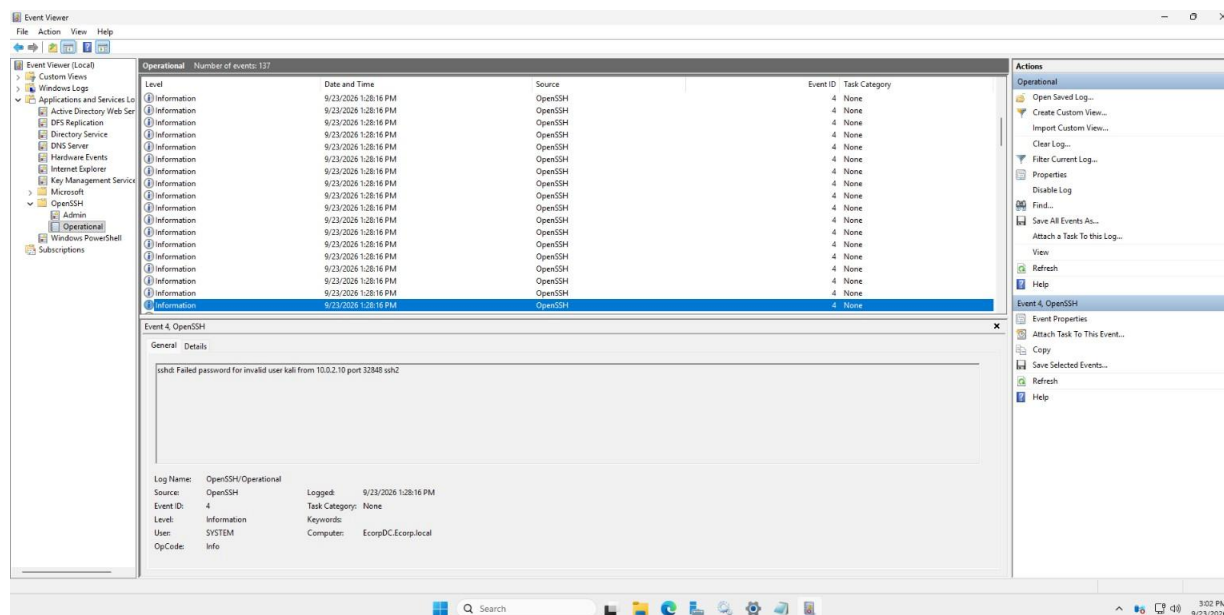


Figure 6: OpenSSH/Operational Event Viewer.

9. Task 6 — Successful Authentication and Compromise Assessment

```
Get-WinEvent -FilterHashtable @{LogName='Security'; Id=4624} -MaxEvents 100 |  
Where-Object {$_.Message -match 'sshtest'} |  
Select-Object TimeCreated, Id, Message | Format-List
```

Event ID 4624 was reviewed to validate successful authentication for sshtest.

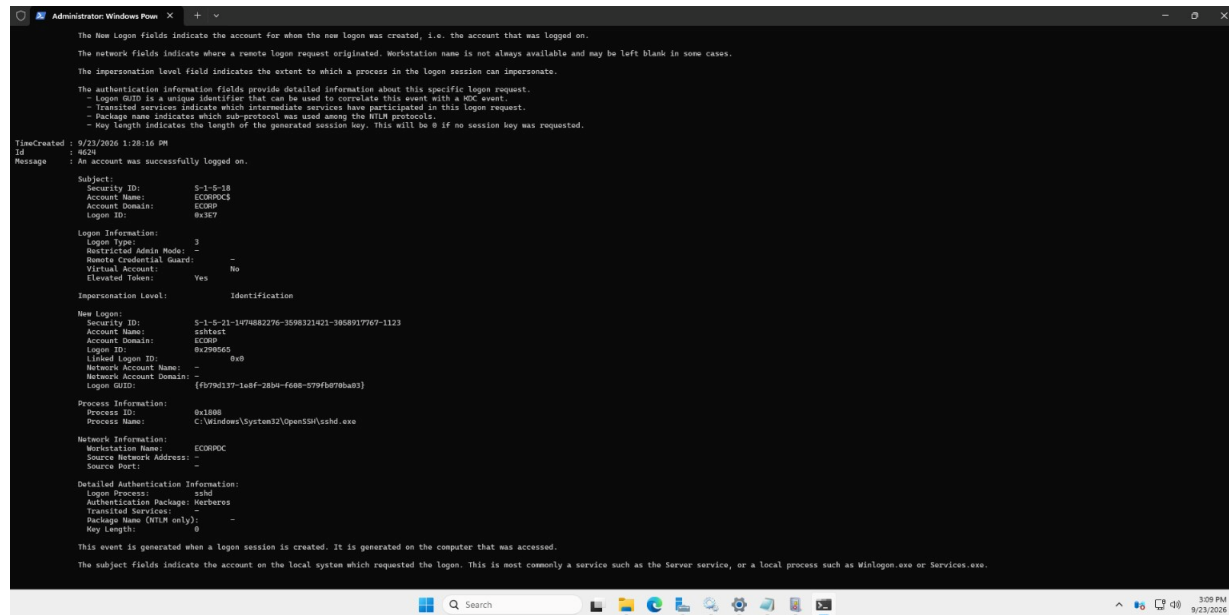


Figure 7: Windows Security Event ID 4624.

10. Task 7 — SSH Hardening and Access Control

Group-based SSH authorization and Windows Defender Firewall source restriction are evidenced. The MaxAuthTries line is shown commented, so it is treated as a recommended additional control rather than an implemented one.

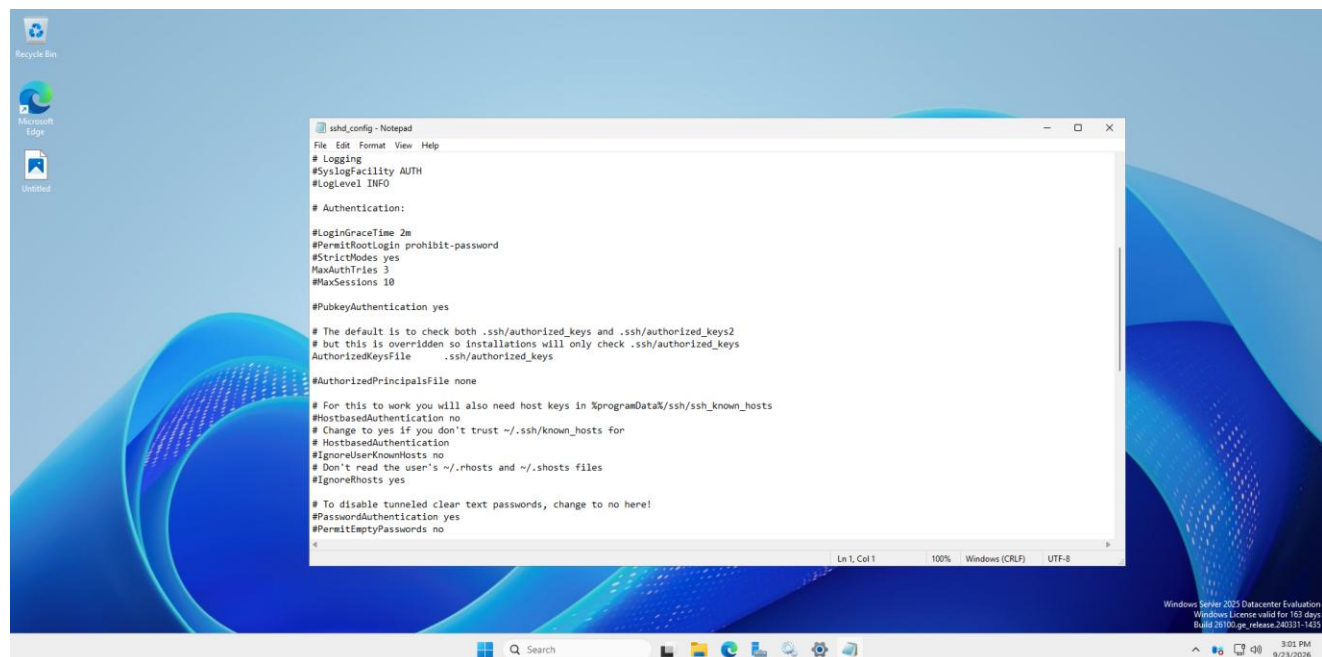


Figure 8: MaxAuthTries Restriction.

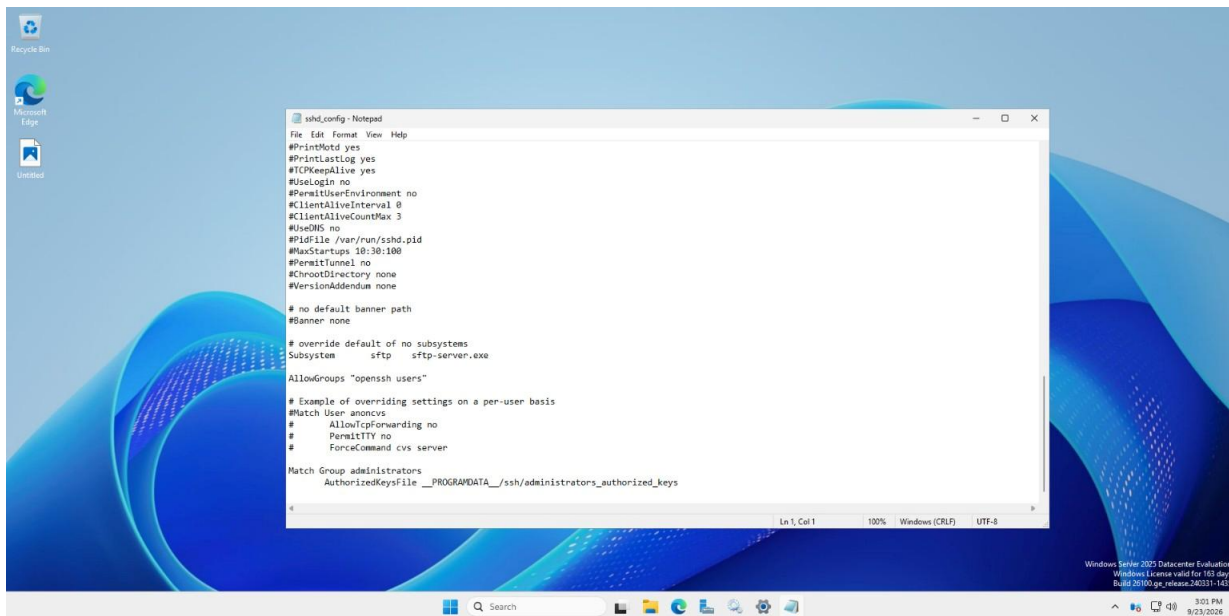


Figure 8: AllowGroups restriction.

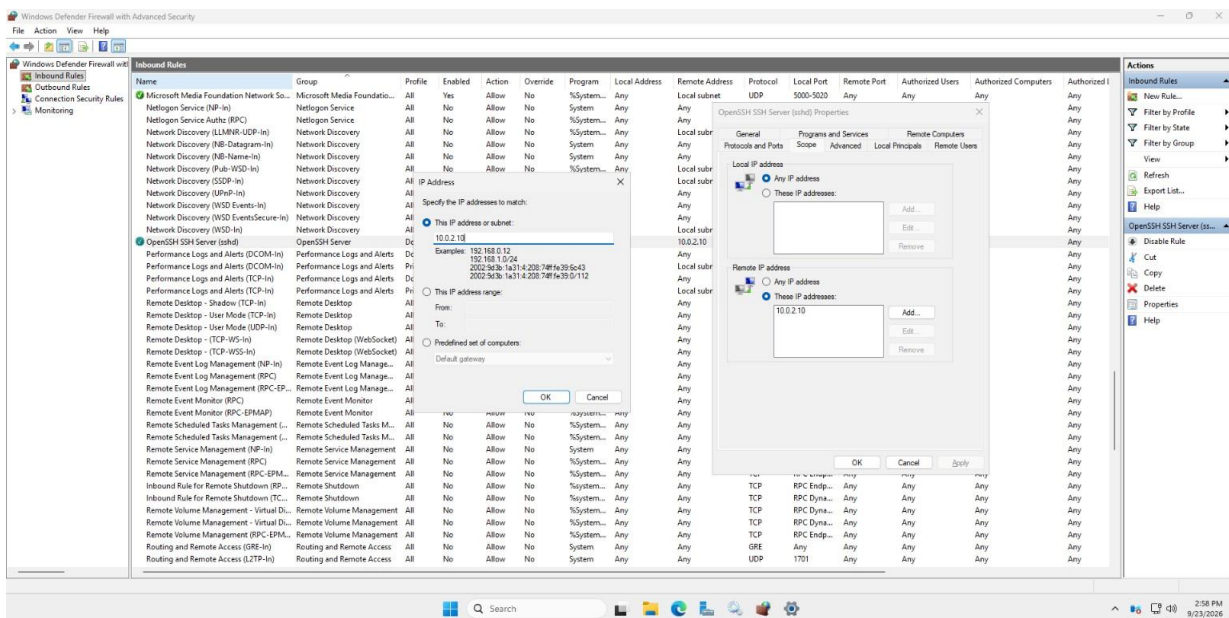


Figure 9: Windows Defender Firewall OpenSSH scope.

11. Task 8 — Public-Key / Certificate-Based SSH Authentication

An Ed25519 SSH key pair was used. The public key was installed through `authorized_keys` and a key-based SSH session was successfully established. This directly demonstrates SSH public-key authentication; no CA-issued certificate is claimed where one is not shown.

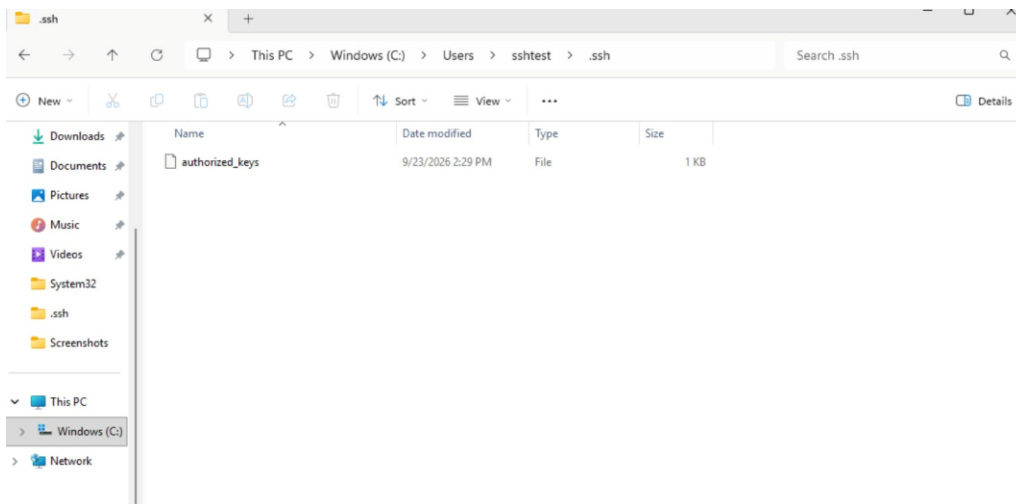


Figure 10: authorized_keys on Windows.

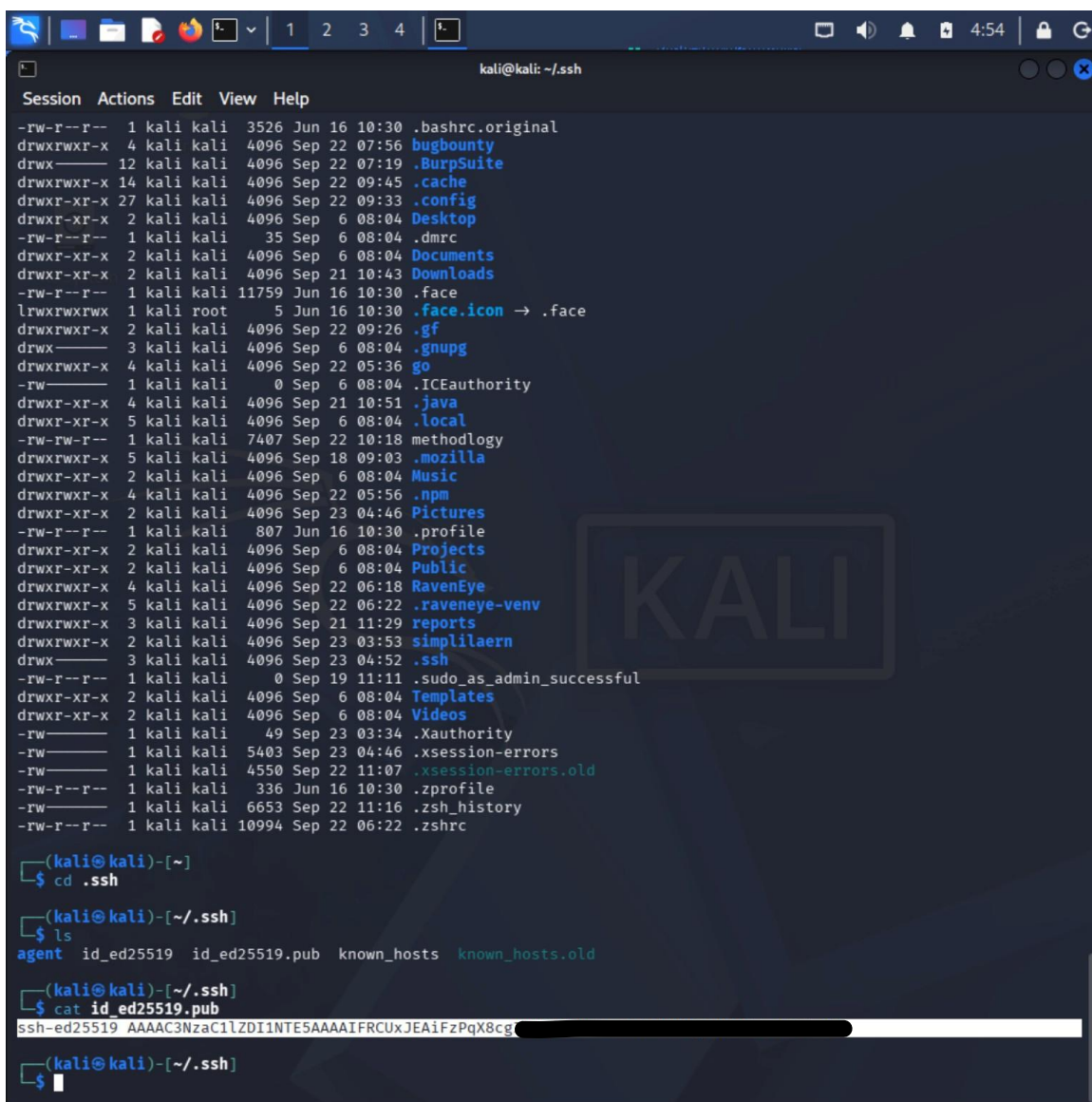
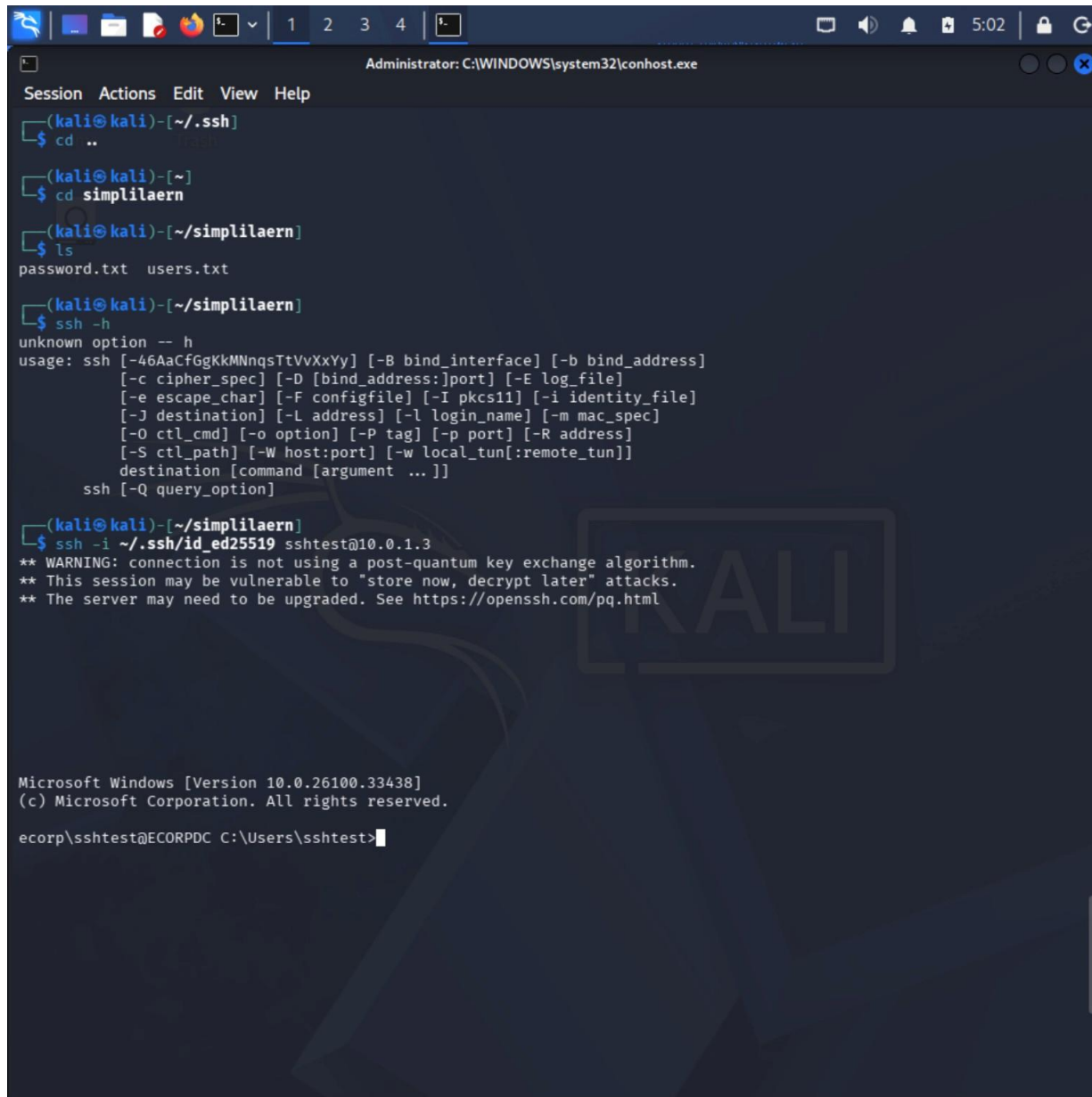


Figure 11: Ed25519 public key on Kali.

```
ssh -i ~/.ssh/id_ed25519 sstest@10.0.1.3
```



The screenshot shows a Windows command prompt window titled "Administrator: C:\WINDOWS\system32\conhost.exe". The terminal session starts in a Kali Linux environment, where the user navigates to the directory `~/ssh` and then `~/simplilaern`. They list files, showing `password.txt` and `users.txt`. Then, they run `ssh -h`, which displays the SSH command-line options. Finally, they execute `ssh -i ~/.ssh/id_ed25519 sstest@10.0.1.3`. The terminal shows a warning about the lack of post-quantum key exchange algorithms, followed by a successful login. The prompt changes to `ecorp\sstest@ECORPDC C:\Users\sstest>`. A large "KALI" watermark is visible in the background of the terminal window.

```
Administrator: C:\WINDOWS\system32\conhost.exe
Session Actions Edit View Help
(kali㉿kali)-[~/ssh]
$ cd ..
(kali㉿kali)-[~/]
$ cd simplilaern
(kali㉿kali)-[~/simplilaern]
$ ls
password.txt  users.txt
(kali㉿kali)-[~/simplilaern]
$ ssh -h
unknown option -- h
usage: ssh [-46AaCfGgKkMnNqSsTtVvXxYy] [-B bind_interface] [-b bind_address]
          [-c cipher_spec] [-D [bind_address:]port] [-E log_file]
          [-e escape_char] [-F configfile] [-I pkcs11] [-i identity_file]
          [-J destination] [-L address] [-l login_name] [-m mac_spec]
          [-O ctl_cmd] [-o option] [-P tag] [-p port] [-R address]
          [-S ctl_path] [-W host:port] [-w local_tun[:remote_tun]]
          destination [command [argument ...]]
          ssh [-Q query_option]
(kali㉿kali)-[~/simplilaern]
$ ssh -i ~/.ssh/id_ed25519 sstest@10.0.1.3
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html

Microsoft Windows [Version 10.0.26100.33438]
(c) Microsoft Corporation. All rights reserved.

ecorp\sstest@ECORPDC C:\Users\sstest>
```

Figure 12: Successful public-key SSH login.

12. Findings and Recommendations

- Monitor OpenSSH/Operational and Windows Security logs.
- Use least-privilege group-based SSH authorization.
- Restrict SSH at the Windows Defender Firewall layer.
- Enable an appropriate MaxAuthTries value after validation.
- Prefer public-key authentication for sensitive SSH access.
- Never submit passwords or private SSH keys.

13. Conclusion

The project reproduced the SSH brute-force scenario in an isolated VMware lab and followed the required detection-and-response workflow: attack generation, log analysis, successful-authentication validation, access-control hardening, and public-key authentication. Credentials are redacted from the report and private keys are excluded.

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